

Card Counting and Kelly Sizing in Blackjack: A Technical Note

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Why This Project Exists

Blackjack is one of the only widely available casino games where a player's odds genuinely depend on information they can track themselves: which cards have already been dealt. This project simulates that fact directly. It builds a model of a six-deck blackjack shoe, plays out simplified basic strategy against it, tracks the count the way a real player would, and compares three ways of deciding how much to bet on each hand. The aim is to make the actual mechanics of card counting, and the actual risk of betting aggressively on a small edge, visible and quantifiable, grounded in the same mathematics that mathematician and hedge fund manager Edward O. Thorp used to first demonstrate that blackjack could be beaten [3].

This document explains the ideas behind the simulation in plain terms: why counting cards works at all, what “true count” actually measures, what the Kelly criterion is and why it is both powerful and dangerous, and what the simulation's results say about each. It assumes no background in blackjack or gambling theory.

Why Counting Cards Works

In a single freshly shuffled deck, the player's odds in blackjack are close to even, with a small edge toward the house, mostly because the dealer wins automatically on any tie where the dealer also has a strong hand, and because the player must commit to hitting or standing before seeing how the dealer's hand resolves. Basic strategy, a fixed set of rules for when to hit, stand, double, or split, was first derived in closed form by Baldwin, Cantey, Maisel, and McDermott using desk calculators in the 1950s [1], and later confirmed and refined by Thorp using computer simulation [3]. Under typical six-deck rules, basic strategy narrows the house edge down to roughly half a percent. That edge is small but real, and it is fixed: it does not change from hand to hand under normal play.

What does change, hand to hand, is the composition of the cards left in the shoe. A deck is dealt without replacement: once a card is shown, it cannot reappear until the shoe is shuffled again. If the cards already dealt happen to be disproportionately low cards (2 through 6), then the cards remaining in the shoe are disproportionately high cards (10s, face cards, and aces). That matters because a deck rich in high cards is better for the player than for the dealer, for two main reasons. The player is paid 3:2 on a natural blackjack (an ace plus a 10-value card), so more aces and 10s remaining means more chances at that bonus payout. And the dealer, who is forced by fixed rules to keep hitting until reaching 17, is more likely to bust when the remaining

deck is full of high cards, since busting becomes more likely the higher the dealer's hand climbs past 11 or 12.

Card counting is a way of tracking, approximately, whether the remaining shoe is currently rich in high cards or low cards, without having to remember every card individually. The system used in this project, Hi-Lo, is one of the point-count systems descended from Thorp's original card-counting work [3] and assigns a value to every card as it is dealt: low cards (2 through 6) are worth +1, middle cards (7 through 9) are worth 0, and high cards (10 through ace) are worth -1. Adding up these values as cards are dealt gives the *running count*. A positive running count means more low cards have been dealt than high cards, which means the remaining shoe is comparatively rich in high cards, good for the player. A negative running count means the opposite.

The running count's significance depends on how much of the shoe is left, so it needs an adjustment before it's useful for betting decisions. A running count of +8 means something very different with five decks left in the shoe than it does with half a deck left. The *true count* corrects for this by dividing the running count by the number of decks remaining:

$$\text{true count} = \frac{\text{running count}}{\text{decks remaining}}$$

True count is the number that actually predicts the player's edge on the next hand, and it is the number this simulation uses to decide bet sizes.

The Three Betting Strategies

Knowing the count only matters if it changes what you do. This project compares three approaches to sizing bets, all played against the same simulated shoes and the same simplified strategy rules, so that the comparison isolates the effect of bet sizing itself.

Flat betting wagers the same fixed amount, \$10, on every hand regardless of the count. This is the control: a player who tracks no information at all and bets the same way whether the deck favors them or not.

Spread betting scales the wager up as the true count rises, with a floor at the \$10 base bet and a cap at eight times that, and a hard \$500 table maximum on top of that. This is closer to how a real, moderately disciplined counter bets: more when the deck is favorable, the table minimum when it is not.

Kelly sizing is a more mathematically aggressive approach, named after a betting formula published by Bell Labs researcher John L. Kelly Jr. in 1956 [2]. Kelly's formula says the optimal fraction of a bankroll to wager on a favorable bet is proportional to the size of the edge, and inversely proportional to the variance of the outcome. Thorp was the first to apply this formula to blackjack betting specifically, recognizing that a card counter's fluctuating edge by true count made it a natural fit for Kelly's framework [3]. This simulation uses a simplified version: it estimates the player's edge as roughly 0.5% per true count, sits out entirely at zero or negative counts (since there's no positive edge to size a bet around), and otherwise bets a fraction of current bankroll based on that estimated edge, subject to the same \$500 table maximum as spread betting.

The Kelly criterion's defining property, and the reason it's worth simulating directly rather than only describing in the abstract, is that it bets as a fraction of *current* bankroll rather than a fixed dollar amount. On a real edge, this compounds: as the bankroll grows, bet sizes grow with it, which produces faster long-run growth than flat or proportionally-capped spread betting, but also means a string of losses costs more in dollar terms once the bankroll has grown. Kelly's original result shows that this approach maximizes the long-run exponential growth rate of capital [2]; it makes no claim about minimizing risk along the way, and the simulation's results below show exactly that tradeoff playing out.

What the Simulation Actually Does

The simulation plays out a six-deck shoe (reshuffled once 75% of it has been dealt) using a deliberately simplified version of basic strategy. It covers the most common hit, stand, double, and split decisions correctly, but it omits surrender, re-splitting, soft doubling, and some upcard-specific nuance that a complete basic-strategy table would include. This was a deliberate scope decision, made to keep the focus on how the count affects outcomes; a simplified-but-honestly-labeled strategy keeps that focus clear without the added complexity of an exhaustive strategy engine.

Results are computed at four checkpoints (1,000, 10,000, 100,000, and 1,000,000 simulated hands), run once offline and stored as precomputed data; the page loads that stored data directly instead of running the simulation live in the browser. The edge-versus-true-count chart on the project page uses only the flat-betting strategy's hands, specifically so that the curve reflects the game's underlying edge at each count, uncontaminated by the fact that spread and Kelly strategies choose to bet very different amounts depending on the count. That chart also displays a 95% confidence interval around each point, which widens noticeably at extreme true counts. This reflects the fact that extreme counts are rare: across a million simulated hands, a true count of +10 occurs in fewer than a thousand of them, so the estimate of edge at that count is necessarily less precise than the estimate at true count 0, where tens of thousands of hands' worth of data are available.

Results

Aggregated across flat-betting hands, the underlying game shows a small negative edge for the player at neutral and negative true counts, roughly -0.7% at true count 0, worsening in deeply negative territory. Edge turns meaningfully positive somewhere around true count +3 to +5, though the confidence band at +3 still spans zero, and even at +4 the interval is wide enough that a single real session at that count could easily look unprofitable despite a genuinely positive long-run edge. That uncertainty reflects a real feature of the underlying data: high true counts are rare, so the data available to estimate the edge at those counts is thinner exactly where the edge matters most.

Over short runs, 1,000 or 10,000 hands, any of the three strategies can finish ahead or behind, because variance from ordinary card-to-card luck dominates at that scale; a thousand hands simply is not enough to reliably separate skill from noise. At 100,000 and 1,000,000 hands, a clearer pattern appears. Flat betting went bust at both long-run checkpoints, despite the underlying game only carrying a modest negative edge, because a fixed wager keeps the player

fully exposed even when the count is unfavorable. Spread and Kelly sizing, by contrast, sit out or bet near the minimum when the count is bad and press their bets when it is favorable, which is precisely what lets them realize a positive edge in dollar terms over the long run, while flat betting cannot.

That upside comes at a real cost in variance. At the million-hand checkpoint, Kelly sizing finished with a bankroll near \$1.38 million, by far the best outcome of the three strategies, but experienced a drawdown of roughly \$167,000 along the way. Spread betting finished more modestly, around \$24,000, with a drawdown near \$30,000. Flat betting, again, went bust. The pattern is consistent with what the Kelly criterion is actually built to do: maximize long-run growth rate on a real edge, at the cost of a much rougher path to get there. In shorter samples, this cuts the other way too. Because Kelly sizes bets as a fraction of a growing bankroll, an unlucky stretch early in a short run can leave it well below its starting point even when the underlying edge is genuinely positive. Card counting shifts the odds in the player's favor at favorable counts. It does not eliminate variance, and a betting strategy built to exploit that edge aggressively will, by construction, also amplify the swings along the way.

What This Project Does and Doesn't Claim

This simulation is a simplified mathematical model of blackjack, not a representation of any specific casino. It uses a simplified strategy table, a simplified linear approximation for how edge scales with true count, and a fixed set of house rules (six decks, 75% penetration, a flat \$500 table maximum) that may not match any specific real casino. It says nothing about whether card counting is practical or legal in a given jurisdiction, nothing about real-world casino countermeasures, and nothing about anyone's actual gambling outcomes. Its purpose is educational: to make the underlying probability, the logic of count-based betting, and the real tradeoff between aggressive sizing and variance, concrete and checkable.

References

- [1] Baldwin, R. R., Cantey, W. E., Maisel, H., & McDermott, J. P. (1956). The Optimum Strategy in Blackjack. *Journal of the American Statistical Association*, 51(275), 429–439.
- [2] Kelly, J. L., Jr. (1956). A New Interpretation of Information Rate. *Bell System Technical Journal*, 35(4), 917–926.
- [3] Thorp, E. O. (1966). *Beat the Dealer: A Winning Strategy for the Game of Twenty-One* (Rev. ed.). Random House.